

Assignment 4 – Business Data Analysis with Deepnote

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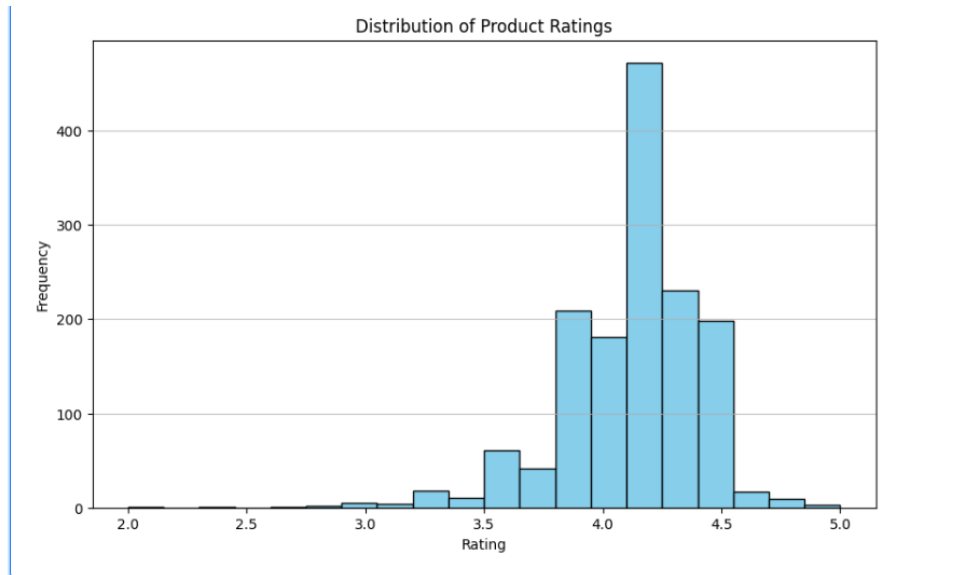
Platform: <https://deepnote.com>

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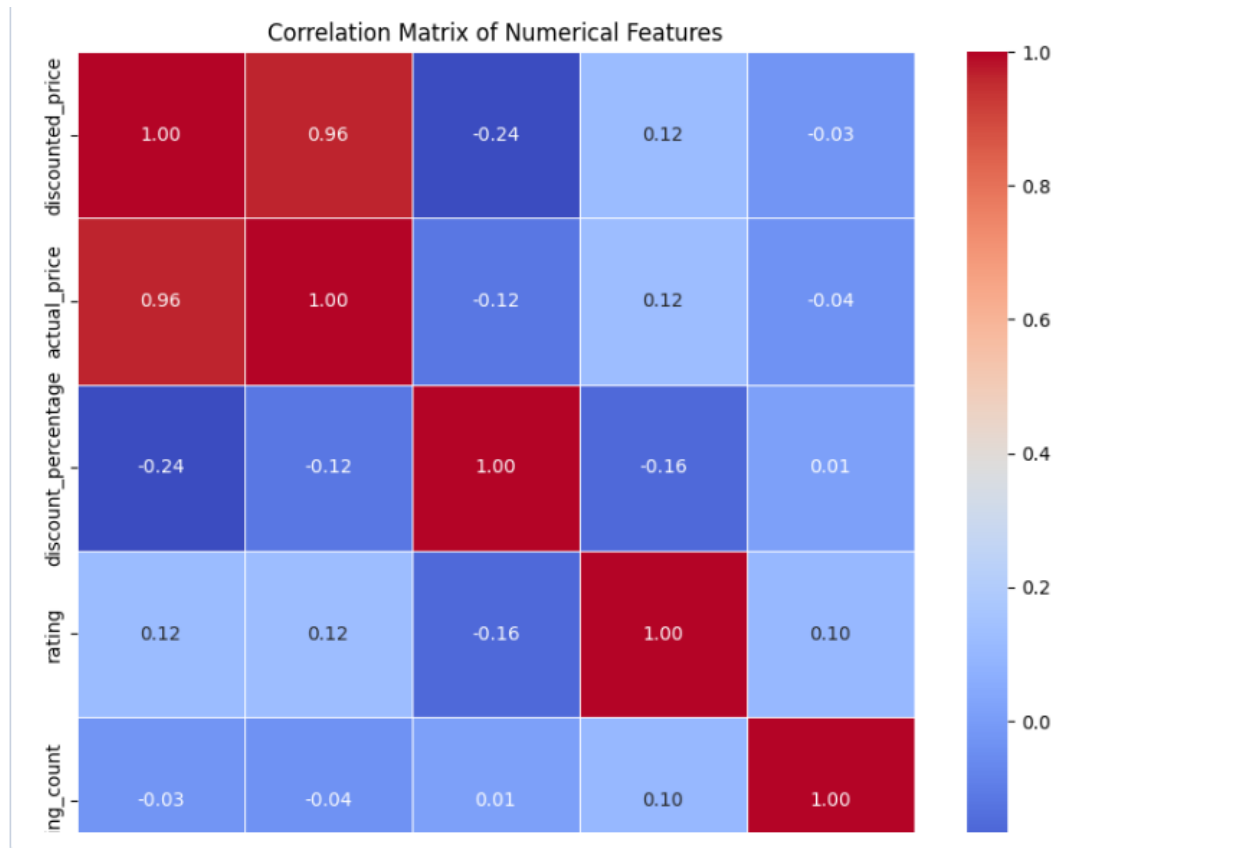
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Lab 1: Explore and Remix a Public Notebook on Deepnote

Notebook link: https://deepnote.com/workspace/Pthao-382c7ef5-1a47-48d2-a84c-c824f158ea6c/project/MIS-395-LAB-1-02519a1b-a5f6-44a3-9178-1446dab3f536/notebook/Notebook-1-38a39ccc8cae4e1595ee1021710237e0?utm_source=share-modal&utm_medium=product-shared-content&utm_campaign=notebook&utm_content=02519a1b-a5f6-44a3-9178-1446dab3f536



→ The majority of products have high ratings, with the most frequent rating being slightly above 4.25. Low ratings are very rare. The distribution is skewed left, indicating a prevalence of positive reviews.



⇒ A product's rating appears to be largely independent of its price and the discount applied. There is no strong evidence to suggest that expensive products are significantly better-rated or that heavily discounted products are significantly worse.

In this lab, I explored a Deepnote notebook analyzing over 1,000 Amazon products, focusing on numerical relationships and review patterns. The correlation matrix showed that discounted and actual prices were highly correlated (0.96), which is expected. Interestingly, product ratings had little to no correlation with price or discount percentage, suggesting that higher or lower prices do not strongly influence customer satisfaction. This challenges the common assumption that expensive products are rated more positively.

The distribution rating chart revealed that most products received high ratings, clustering around 4.0 to 4.5, with very few low scores. This skewed-left distribution indicates a strong tendency for customers to leave positive reviews, possibly due to selection bias or satisfaction with Amazon's curation.

Lab 2: Build Your Own Business Analytics Notebook on Deepnote

Notebook link: https://deepnote.com/workspace/Pthao-382c7ef5-1a47-48d2-a84c-c824f158ea6c/project/LAB-2-MIS-395-06fee085-947c-4aef-87a1-974d4ada2db5/notebook/Notebook-1-8c107b0a017347bd89ede1326a2784ec?utm_source=share-modal&utm_medium=product-shared-content&utm_campaign=notebook&utm_content=06fee085-947c-4aef-87a1-974d4ada2db5

Project title: E-Commerce Shipping Delay Analysis

This project explores a real-world e-commerce dataset to understand factors that influence delivery delays. The goal is to analyze how shipment methods, warehouse performance, and customer behavior relate to whether a product is delivered on time.

Objective:

- Perform exploratory data analysis (EDA)
- Identify key patterns related to delivery delays
- Build a simple predictive model
- Provide data-driven recommendations to improve logistics efficiency

Introduction & Business Question

In the competitive e-commerce landscape, timely delivery is crucial for customer satisfaction and retention. This analysis investigates an e-commerce company's shipping data to address a critical operational problem: a significant portion of shipments are not arriving on time. By exploring various factors related to the products, customers, and logistics, we aim to uncover the root causes of these delays.

Business Question:

- What are the primary factors that correlate with late shipments, and what actionable insights can we derive to improve on-time delivery performance?

Dataset Overview

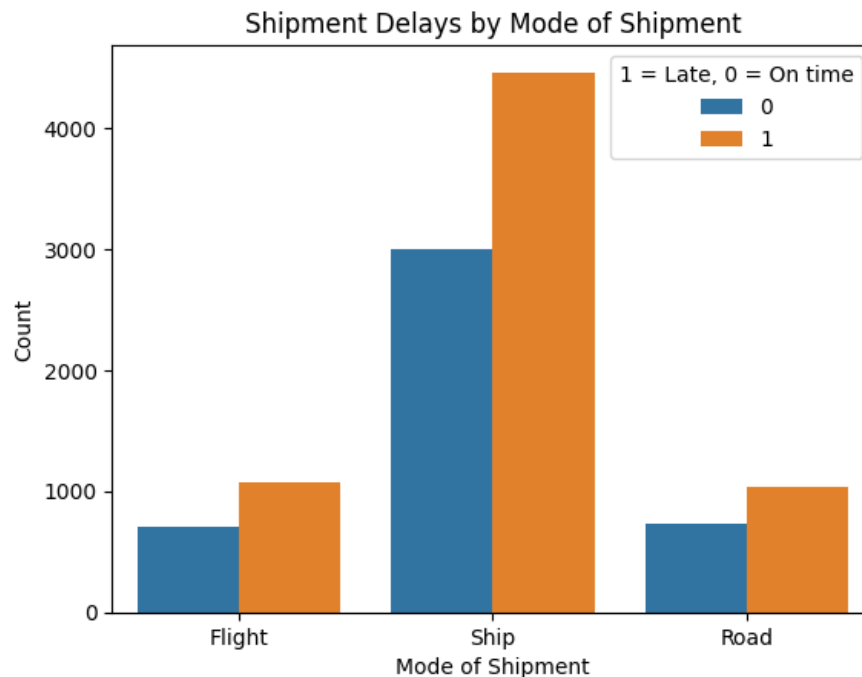
The dataset contains **10,999 records** of individual shipments, with **12 distinct attributes** for each. It provides a comprehensive view of the shipping process, from warehouse logistics to customer interactions.

Key Columns:

- **Logistics Info:** Warehouse_block, Mode_of_Shipment
- **Product Info:** Cost_of_the_Product, Product_importance, Weight_in_gms, Discount_offered
- **Customer Info:** Customer_care_calls, Customer_rating, Prior_purchases, Gender
- **Target Variable:** Reached.on.Time_Y.N (This is our key metric, where 1 indicates a late delivery and 0 indicates an on-time delivery).

An initial check confirms the dataset is complete, with **no missing values**, allowing us to proceed directly to the analysis. The data includes a healthy mix of numerical and categorical features, which is ideal for a detailed exploratory data analysis (EDA).

Shipment Delays by Mode

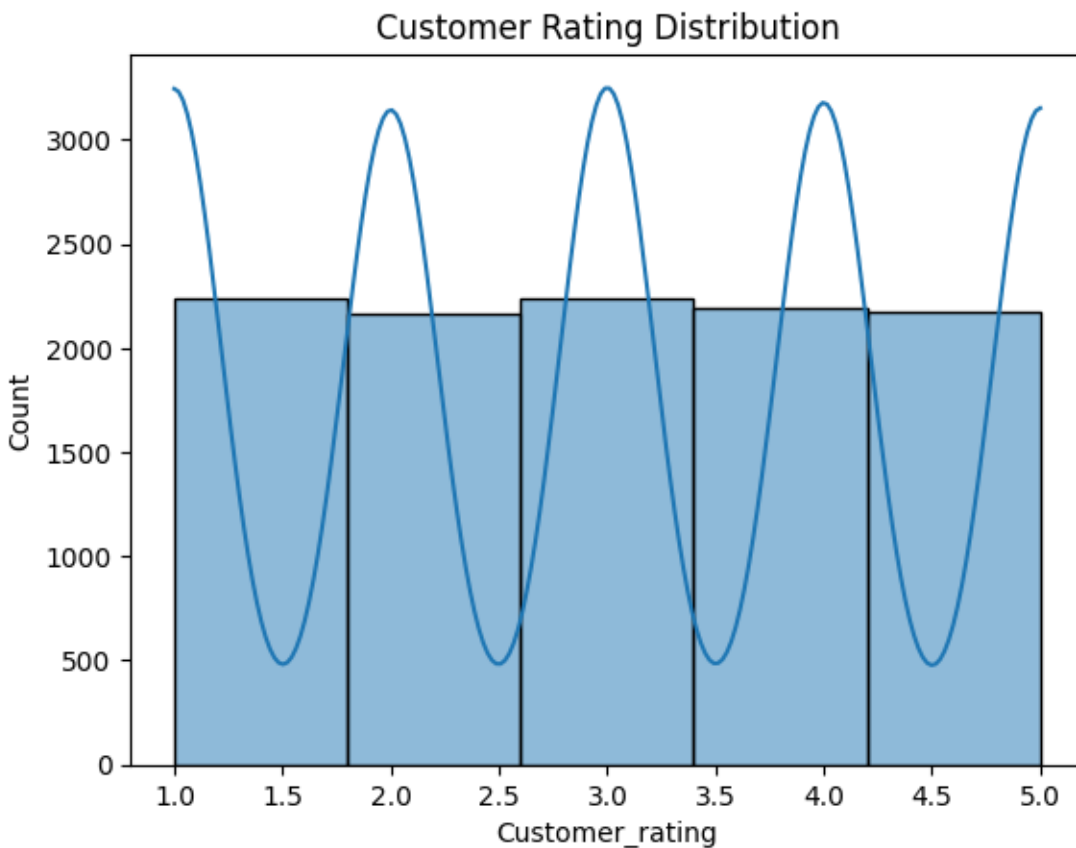


Shipments using the ship mode have the highest number of late deliveries, followed by road.

Flight has the fewest late shipments, indicating it's the most reliable mode in terms of on-time performance.

The company should reconsider its heavy reliance on shipping by sea if on-time delivery is a critical metric.

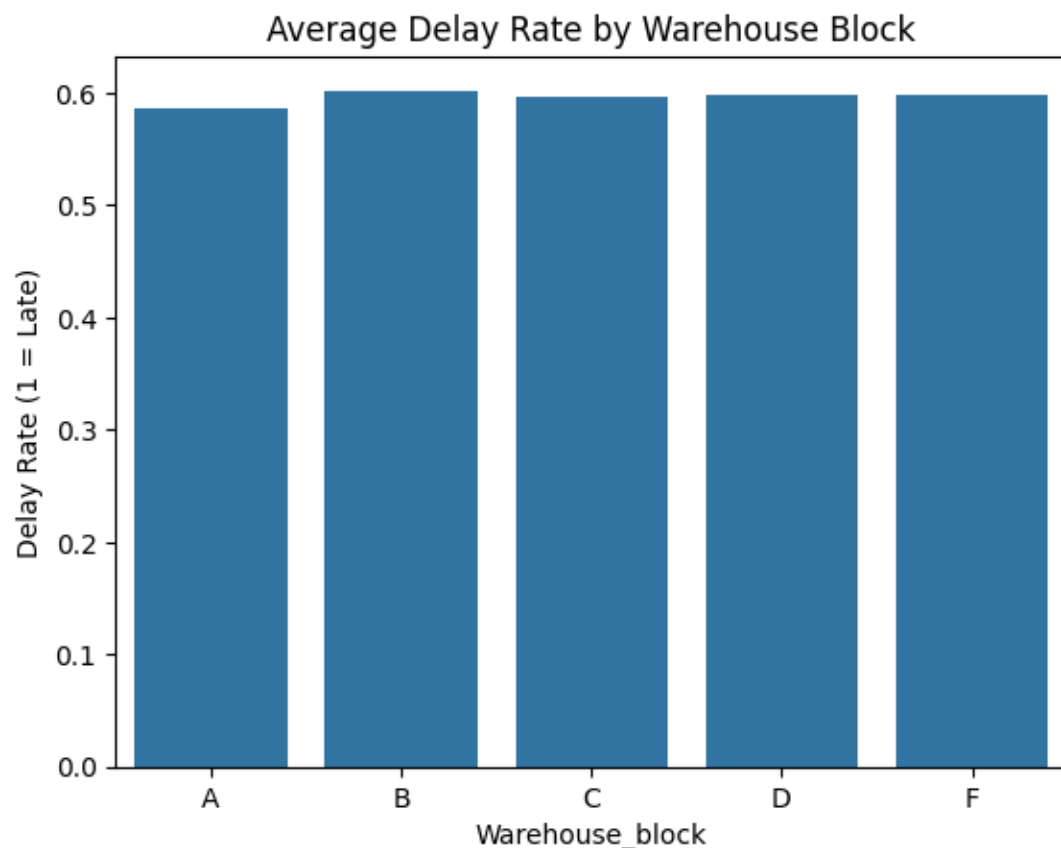
Customer Rating Distribution



The customer ratings are quite evenly distributed across all five rating levels (1 to 5).

This suggests that customer satisfaction is mixed, with a slight concentration around neutral to positive experiences. However, the presence of lower ratings (1–2) still signals room for service improvement.

Delay Rate by Warehouse



All warehouse blocks (A, B, C, D, F) have similar average delay rates (around 0.59 to 0.60), with no extreme outliers.

This suggests that warehouse performance may not be the main factor influencing delays — the issues likely lie in the shipping method or other operational aspects.

Build Simple Predictive Model

	precision	recall	f1-score	support
0	0.55	0.57	0.56	895
1	0.69	0.68	0.69	1305
accuracy			0.63	2200
macro avg	0.62	0.62	0.62	2200
weighted avg	0.63	0.63	0.63	2200

The logistic regression model achieves an accuracy of 63%, with a precision of 0.69 for detecting late shipments (class 1).

While the model does moderately well at predicting delays, there is still room for improvement — potentially by including categorical variables (e.g., one-hot encoding for Warehouse_block, Mode_of_Shipment, etc.).

Conclusion & Business Implications

The exploratory analysis shows that shipping method is the strongest factor affecting delivery timeliness, with shipments by sea experiencing the most delays. Customer satisfaction is distributed evenly, though there is a noticeable share of low ratings that may correlate with shipment issues.

Warehouses show similar delay rates, indicating that warehousing is not the key cause of late deliveries. The company should consider optimizing its logistics strategy, including:

- Promoting flight shipping for high-priority or time-sensitive products
- Reassessing the reliability of ship transport
- Enhancing customer communication for delayed deliveries to mitigate dissatisfaction
- Investing in predictive models to identify delay risk in advance

Limitations and Future Improvements

- Categorical variables were not encoded (e.g., 'Warehouse_block', 'Product_importance', 'Mode_of_Shipment') - adding them may improve model accuracy.
- Feature engineering was limited; additional derived metrics (like shipping distance or customer type) could improve insights.
- External factors such as weather, region, or traffic conditions were not included.
- Future work could explore classification models (e.g., decision trees, random forest) and deploy delay prediction dashboards for real-time logistics monitoring.